ELECTRONICS

Prepared by Eng. Manar Ahmed

✓ Section outlines

- Breadboard Implementation of Circuit -1.
- Circuit-2.1 ⟶ Control LED Intensity using a Potentiometer.
- Circuit -2.2 ⟶ Slow Light Bulb .
- Assignment .

Circuit-2.1 Control LED Intensity using a Potentiometer.

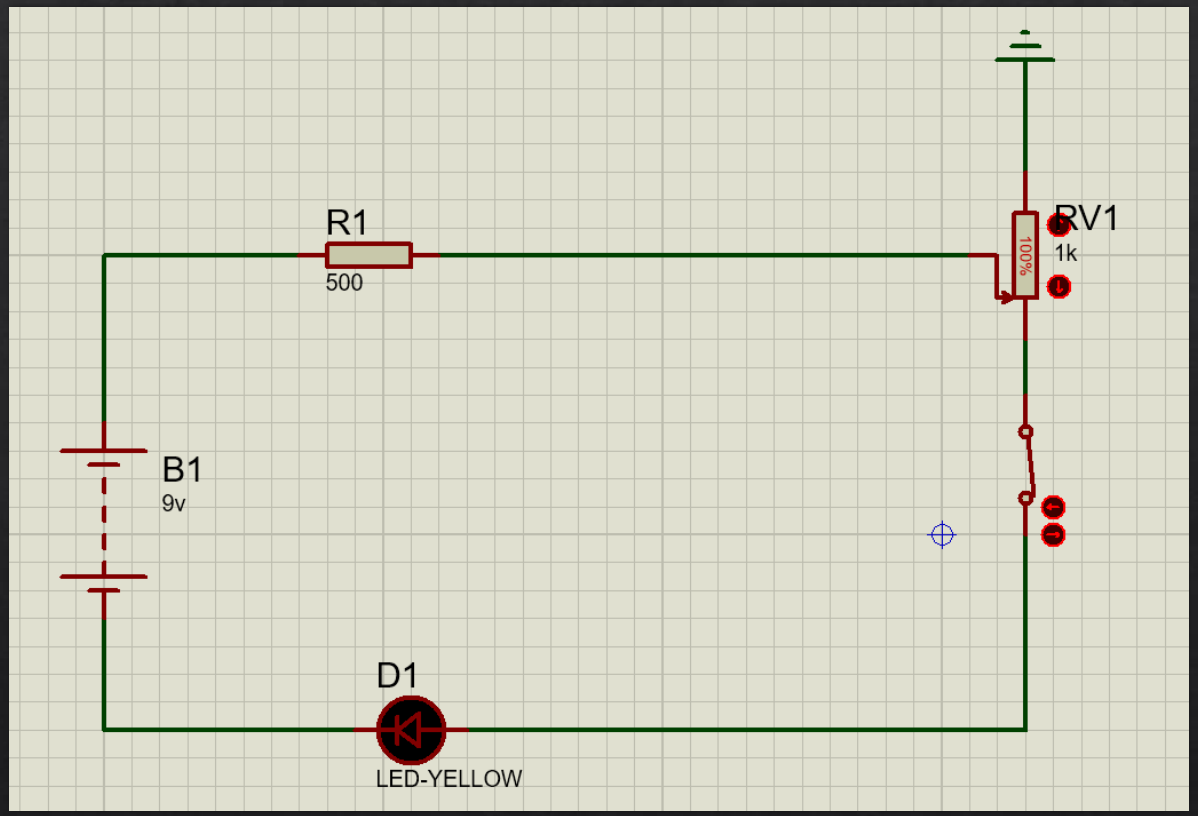
Circuit 2.1: Control LED Intensity using a Potentiometer

✓ **objective:**

•Understand how to control LED brightness.
•Learn how a potentiometer works in a circuit.
•Observe how resistance affects current and light intensity.

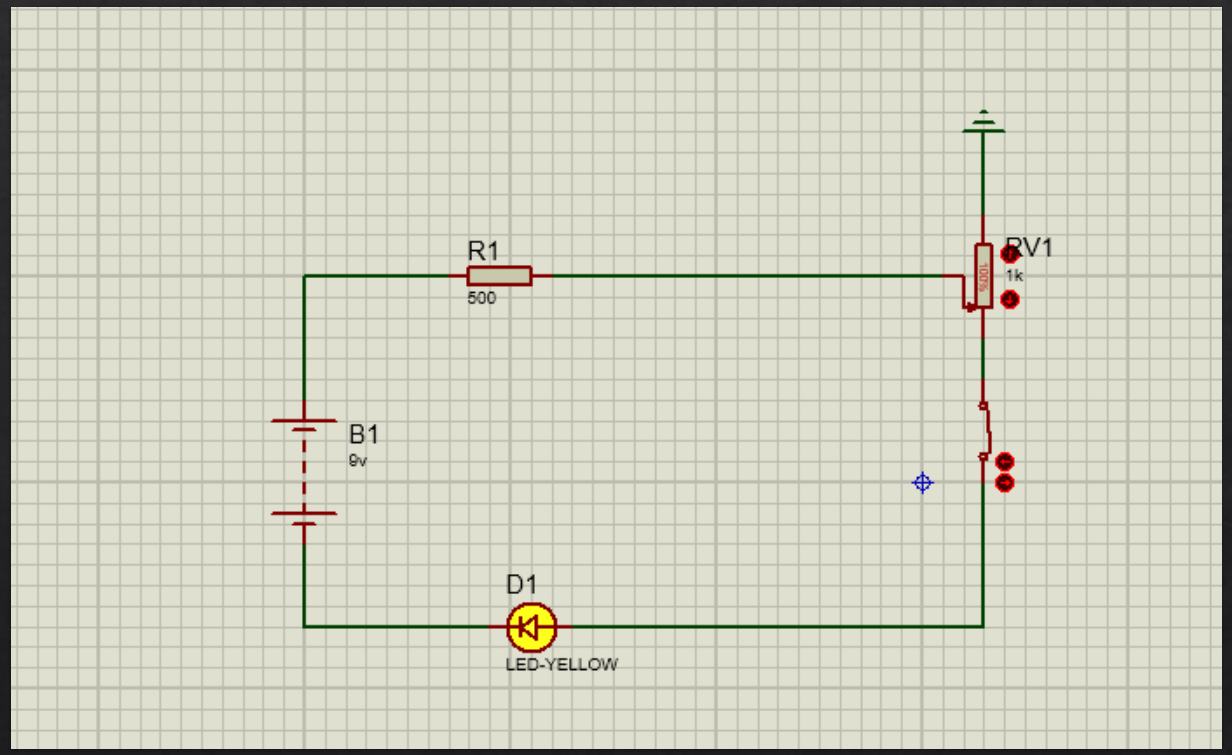
✓ **Components Used:**

•9V Battery
•Switch
•Resistor (500Ω)
•Potentiometer (1kΩ)
•LED



Role of Each Component

- ✓ **Battery** ⟶ Provides electrical power (9V).
- ✓ **Switch** ⟶ Turns the circuit ON and OFF.
- ✓ **Resistor (R1 = 500Ω)** ⟶
  1-Protects the LED from excessive current.
- ✓ **Potentiometer (RV1 = 1kΩ)**
  1-Adjusts the resistance.
  2-Controls LED brightness.
- ✓ **LED** ⟶ Emits light when current flows through it.



Observation

- ✓ Changing the potentiometer position changes LED brightness.
- ✓ Higher resistance → lower current → dim LED.
- ✓ Lower resistance → higher current → bright LED.

Circuit -2.2 Slow Light Bulb .

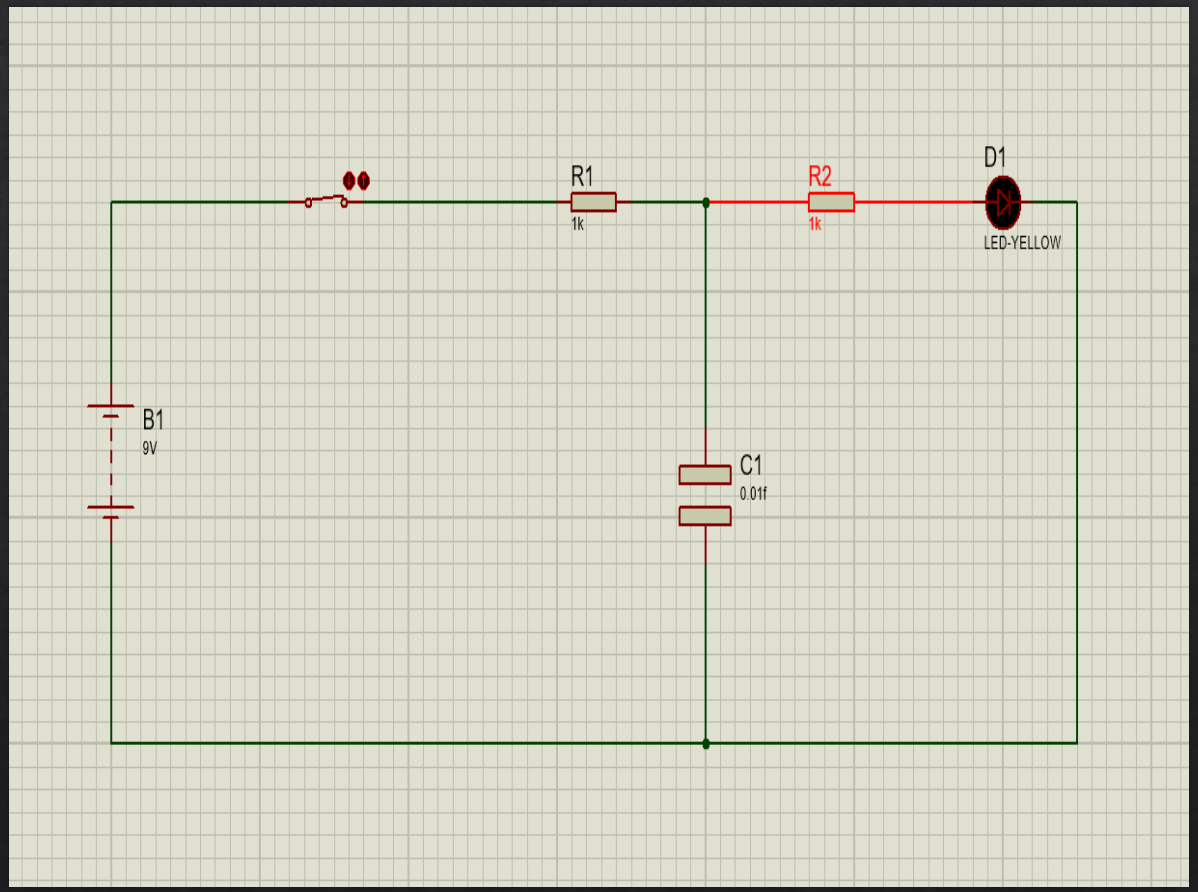
C -2.2 Slow Light Bulb

✓ **objective:**

•Understand how a capacitor affects LED lighting.
•Learn the concept of charging and discharging.
•Observe how a capacitor can create a delay effect.

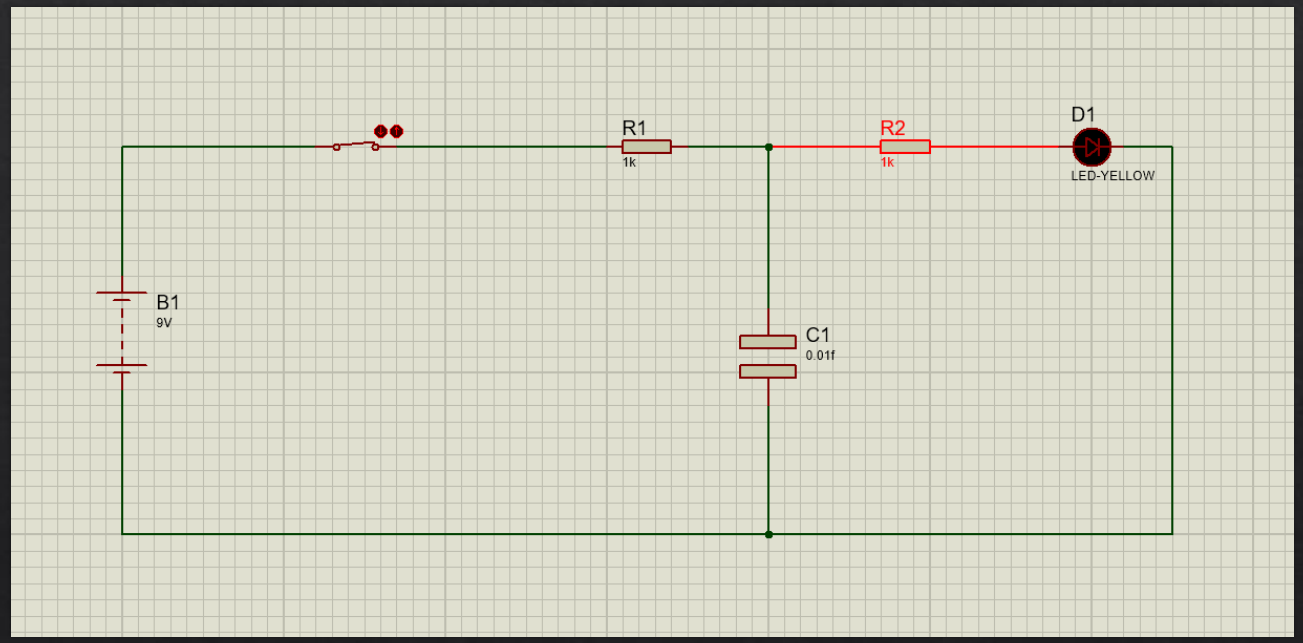
✓ **Components Used:**

•9V Battery
•Switch
•Resistor R1 (1kΩ)
•Resistor R2 (1kΩ)
•Capacitor C1 (0.01F)
•LED



Role of Each Component

- ✓ **Battery** ⟶ Supplies electrical energy to the circuit.
- ✓ **Switch** ⟶ Starts or stops the circuit.
- ✓ **Resistors (R1 & R2)** ⟶ Limit current to protect the LED.
- ✓ **Capacitor (C1)** ⟶ Stores electrical charge temporarily.
- ✓ **LED** ⟶ Converts electrical energy into light



Working Principle (Charging)

- When the **switch is turned ON**:

1. Current starts flowing from the battery.
2. The capacitor begins charging.
3. Voltage across the circuit increases gradually.
4. The LED turns on slowly instead of instantly.

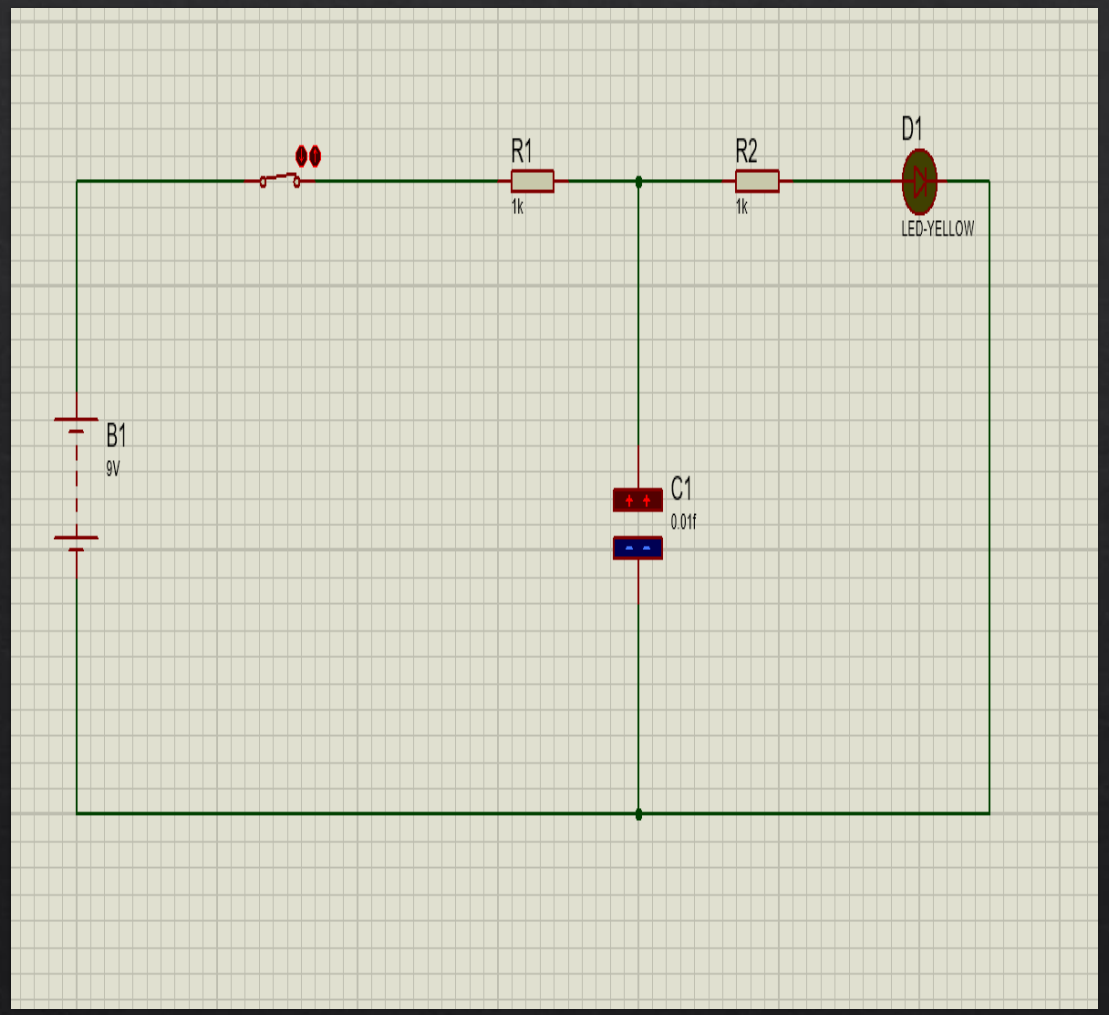
- This happens because the capacitor takes time to charge.

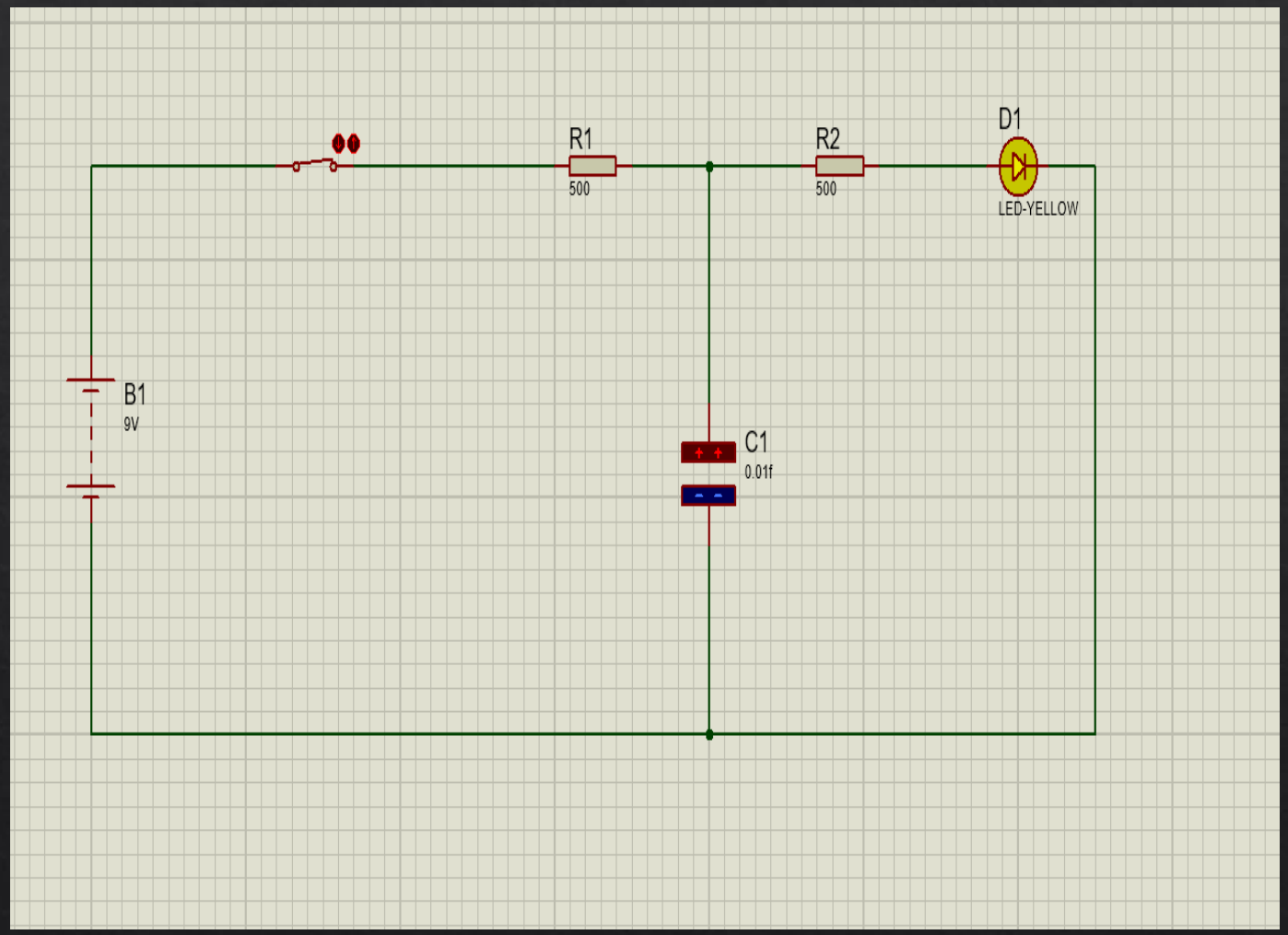
Working Principle(Discharging)

- When the switch is turned OFF:
- The capacitor still holds stored charge.
- It begins discharging through the circuit.
- The LED fades slowly instead of turning off immediately.
- This creates the slow light effect..

- This creates the **slow light effect**.

Circuit Diagram





✓ Assignment

- Breadboard Implementation of Circuit -2.2 .
- software implementation of Circuit -2.2 .
- software implementation of Circuit -2.1 .
- Breadboard Implementation of Circuit -2.1 (video) .

Thanks

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